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Immersion cooling using cooling containers for an electronic device

Abstract

An electronic device includes a chassis, electronic components disposed within the chassis, cooling containers, and a coolant loop. The chassis has an internal volume partly filled with coolant to form coolant pool in the internal volume. The containers are disposed within the chassis such that each container contains at least one electronic component. The coolant loop circulates the coolant from the coolant pool to the containers. For example, each container receives an inflow of the coolant from the coolant loop or from another one container and immerse the at least one electronic component in the coolant to remove heat from the at least one electronic component and output an outflow of the coolant that overflows into the coolant pool or into at least one adjacent container, where a level of the coolant in the coolant pool is lower than respective levels of the coolant in the containers.

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Background/Summary

BACKGROUND

(1) Datacenters typically include racks, where each rack has multiple electronic devices, which are configured to execute workloads. Generally, some electronic components (e.g., a central processing unit (CPU), graphics processing unit (GPU), or the like) of such electronic devices may consume high amounts of power, thereby increasing the power density of the rack and the amount of heat generated from such electronic devices. Cooling techniques, such as air cooling techniques are typically deployed to remove heat from the electronic devices. However, since air does not have good heat transfer capabilities, the air cooling techniques may not be very efficient in removing heat from the electronic devices. Further, as the power of electronic devices increases, air cooling techniques may require additional power to sufficiently cool the electronic devices, resulting in a higher power density of the rack. Hence, a new cooling technique, such as immersion cooling may be deployed in the datacenter, which may use liquid coolant instead of air to remove heat from the electronic devices. By utilizing the superior heat transfer capabilities of the liquid coolant compared to air, electronic devices may be cooled more efficiently, thus reducing the overall power usage of the datacenter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Various examples will be described below with reference to the following figures.
- (2) FIG. 1A illustrates a top view of an electronic device according to an example of the present disclosure.
- (3) FIG. 1B illustrates a side view of the electronic device of FIG. 1A illustrated while viewing from direction “A” in FIG. 1A according to an example of the present disclosure.
- (4) FIG. 1C illustrates another side view of the electronic device of FIG. 1A illustrated while viewing from direction “A” in FIG. 1A according to the example of the present disclosure.
- (5) FIG. 2 illustrates the electronic device **100** of FIG. 1A having a controller and a plurality of sensors according to the example of the present disclosure.
- (6) FIG. 3 illustrates a block diagram of a rack having an electronic device of FIGS. 1A-1C and a facility liquid supply unit according to an example of the present disclosure.
- (7) FIG. 4 is a flowchart depicting a method of cooling a plurality of electronic components and a primary circuit board of an electronic device according to one example of the present disclosure.

DETAILED DESCRIPTION

(8) The following detailed description refers to the accompanying drawings. For purposes of explanation, certain examples are described with reference to the components illustrated in FIGS. 1-4. The functionality of the illustrated components may overlap, however, and may be present in a fewer or greater number of elements and components. Moreover, the disclosed examples may be implemented in various environments and are not limited to the illustrated examples. Wherever possible, the same reference numbers are used in the drawings and the following description to refer to the same or similar parts. It is to be expressly understood, however, that the drawings are for the purpose of illustration and description only. While several examples are described in this document, modifications, adaptations, and other implementations are possible. Accordingly, the following detailed description does not limit the disclosed examples. Instead, the proper scope of the disclosed examples may be defined by the appended claims.

(9) Immersion cooling may require electronic devices to be completely submerged in coolant to remove heat from electronic components of the electronic devices. For example, previously immersion cooling has generally involved providing a tank, a tub, or other large container to hold multiple servers and to be filled with liquid coolant. Such immersion cooling tank may be similar to a traditional rack of the datacenter in terms of their size and the number of servers or other

electronic devices housed therein, but unlike a traditional rack the immersion cooling tank is liquid tight and filled with the coolant such that the electronic devices may be completely submerged in the coolant to remove heat from the electronic components. The immersion cooling may further require periodic circulation of the coolant in the tank to a heat exchanger to remove heat from the coolant and refill the tank with the coolant. Since the tank is completely filled with the coolant, the tank may be very heavy (much heavier than a traditional rack of equivalent size) and may require a large quantity of coolant to submerge the electronic devices. The coolant generally needs to be an electrically non-conductive liquid to avoid damaging the electronic components, and thus the coolant can be expensive, especially when used in large quantities as may be needed for traditional immersion cooling. Further, since the electronic devices need to be completely submerged in the coolant, the tank may have to be oriented with its longest side extending along the floor (e.g., horizontally oriented) to accommodate multiple electronic devices, thereby increasing floor space requirements in the datacenter to deploy the tank (as compared to traditional racks, which are generally oriented with the longest side extending vertically). Since traditional racks typically extend along its height (e.g., vertically oriented), for implementing such immersion cooling, the datacenter may have to include a new infrastructure (e.g., a horizontally oriented tank and also a new coolant distribution unit (CDU)). Therefore, the existing infrastructure (e.g., vertically oriented traditional rack) cannot be reused. Additionally, since all the electronic devices are completely submerged in the same liquid coolant filled in the tank, the high and low-power consuming electronic components of all such electronic devices are provided with the same coolant, which may result in poor heat dissipation from such high-powered electronic components, and also may necessitate frequent circulation of the coolant from the tank to heat exchanger for removal of heat from the coolant. Additionally, since the electronic devices are completely submerged in the liquid coolant in the tank, it can be difficult to service (e.g., add, remove, replace, or perform maintenance on) individual electronic components (e.g., storage drives, PCIe add-on cards, memory, processors, etc.) or to change how communication cables are coupled to the electronic device, because such actions may require removing the entire electronic device from the tank. This removal of the entire electronic device from the tank can be messy, since the device is submerged, and thus servicing may be more complicated, difficult, and time consuming than it would be to perform a similar service on an electronic device in a traditional rack. In addition, the electronic device may be forced to undergo unavoidable shut down during such events (because it is removed entirely from the tank), and therefore hot swapping (adding, removing, or replacing electronic components to the electronic device while the electronic device is still running) may not be possible.

(10) A technical solution to the aforementioned problems includes partially filling individual electronic devices (e.g., servers) with liquid coolant such that the electronic components within the electronic device are submerged but the entire electronic device is not itself submerged (i.e., the coolant is contained within the interior of the chassis of the electronic device). Thus, immersion cooling is provided for the electronic components of the electronic devices on a per-device basis, instead of fully immersing multiple electronic devices in a large tank.

(11) More specifically, in some examples the technical solution includes providing an electronic device that includes a chassis and multiple cooling containers disposed within the chassis, each containing at least one electronic component, where the chassis and the cooling containers are at least partially filled with coolant to immerse the electronic components of the electronic device. The chassis (or a structure disposed therein) defines a volume that can hold coolant (e.g., the volume may be liquid tight on at least a bottom and all lateral sides thereof), and this volume may be partially filled with the coolant to form a coolant pool. The multiple cooling containers may be deployed in the chassis of the electronic device and communicably coupled to the coolant pool such that the coolant can be circulated between the coolant pool and each cooling container. That is, each cooling container may contain at least one electronic component and receive the coolant from the coolant pool via a coolant loop to immerse the at least electronic component with the

coolant to remove heat generated from such electronic component. The heated coolant may then be returned from the cooling containers to the coolant pool, the heated coolant may be cooled (e.g., by a heat exchanger), and then the coolant may be input again into the cooling containers.

(12) In examples disclosed herein, because the entire electronic device is not submerged, the entire electronic device does not need to be completely removed from its environment to allow for service, for example, the electronic device can be serviced while still installed in a rack. Thus, examples disclosed herein may allow for servicing to occur without necessarily requiring shutting down of the electronic device. In addition, the mess and complication associated with removing a large and heavy electronic device from an immersion tank and draining the coolant therefrom can be avoided. Moreover, in some examples, the volume that contains the coolant pool and some or all of the cooling containers may be open on a top side thereof (or have a cover that is removeable on the top side thereof). This can make servicing electronic components even easier, as it can allow a user to access the electronic components within a given cooling container, for example, an electronic component can be added or removed by simply inserting or removing the electronic component through the open top side of the corresponding cooling container. Furthermore, in many cases, such servicing can be performed even while the coolant remains present around the electronic components. In other cases, if it is desired for the coolant to not be present around the electronic components during the servicing, then it may be possible to drain just the cooling container that contains the electronic components being serviced, without necessarily needing to drain the entire electronic device.

(13) In some examples, some or all of the cooling containers are contained within the volume defined in the chassis, and these cooling containers may define sub-compartments within the larger volume. Dividing up the volume into sub compartments via the cooling containers may be beneficial, as compared to just having one volume within the chassis to hold the coolant, at least because the cooling containers may allow for certain electronic components to be fully submersed without requiring as high of a liquid level within the overall volume. For example, if only the one volume were provided and if one or more electronic components are relatively tall compared to the remainder of electronic components, then the liquid level within the volume would need to be relatively high to ensure that the tall electronic component is fully immersed, and this may result in a large amount of coolant being used, which can be costly and make the electronic device very heavy. But if the tall electronic components can be enclosed with smaller compartments, then the liquid level can be high only within the smaller compartments to fully immerse the tall electronic component, while allowing the liquid level within the larger volume to remain relatively low, thus reducing the overall amount of coolant that is used. In addition, by segregating certain electronic components into separate compartments, an order or priority of cooling can be imposed by providing the coolant to some cooling containers prior to (i.e., upstream of) other cooling containers, which allows electronic components that need colder coolant to receive the coolant before it has been heated by other electronic components, and electronic components that can tolerate warmer coolant to receive the coolant after it has removed heat from other electronic components in another cooling container upstream thereof.

(14) In some examples, some of the cooling containers may receive coolant directly from the coolant loop, while others of the cooling containers may receive coolant from another one of the cooling containers. For example, one or more of the cooling containers may be designed to overflow the coolant to the coolant pool or to adjacent cooling containers to circulate the coolant.

(15) In some examples, one or more electronic components of the electronic device are immersed in the coolant pool in the volume (as opposed to being disposed in one of the cooling containers). This allows the coolant to further remove heat from the PCB disposed in the chassis and also from the electronic components located within the cooling container. For example, in some implementations a printed circuit board (PCB) (e.g., a main system board or a motherboard) is disposed within the volume and at least a portion of the PCB is immersed in the coolant pool in the

volume. In some of these examples, some or all of the cooling containers may be coupled to the PCB and may contain electronic components that are mounted to the PCB. In other words, some portions of the PCB are immersed in the coolant pool in the larger volume, while other electronic components mounted to the PCB are segregated into separate compartments via the cooling containers. For example, in some implementations, some of the cooling containers may include lateral walls that define a perimeter laterally encircling at least one electronic component, and the lateral walls are coupled to, and extend perpendicularly from, the face of the PCB upon which the at least one electronic component is mounted such that a portion of the face of the PCB defines a bottom wall of the cooling container. Thus, for example, a portion of the face of the PCB that is located outside of the cooling container may be immersed within the coolant pool, while the portion of the face of the PCB that is located within the cooling container may be immersed within a separate pool contained within the cooling container, wherein the liquid in the cooling container may have a different temperature and/or a different level (height) than the liquid in the coolant pool. Segregating portions of the PCB such that some are immersed within the coolant pool while other portion are immersed within the cooling containers coupled to the PCB can allow for colder coolant to be provided to the electronic components in the cooling containers (such as a processor or CPU), which may generate more heat and/or be more heat sensitive, while warmer coolant is provided to the remainder of the PCB, which may generate less heat and/or be less heat sensitive, which allows for more efficient cooling as noted above.

(16) In some examples, a first cooling container may be deployed containing a high-powered electronic component, such as a CPU or a GPU to receive the coolant directly from the coolant loop, remove heat from such high-powered electronic component. Later, the first cooling container may be fluidically coupled to a second cooling container, which may be deployed containing a low-powered electronic component (e.g., DIMMS) to receive the coolant from the first cooling container. In particular, the first cooling container may be designed to overflow the coolant to the second cooling container. In such examples, the second cooling container may immerse the low-powered electronic component with the coolant received from the first cooling container to remove heat from such low-powered electronic component. Later, the second cooling container may overflow the coolant into the coolant pool. The coolant in the coolant pool may cool portions of a primary circuit board of the electronic device (i.e., the portions not contained within the cooling containers).

(17) The coolant loop may include a pump, which may be fluidically coupled to a coolant collector disposed in the coolant pool via a heat exchanger. In such examples, the pump may draw the coolant through the heat exchanger, causing the heat exchanger to remove heat from the coolant, and provide the coolant to the multiple cooling containers of the electronic device.

(18) In some examples, the pump may be a passive pump, which may be configured to circulate the coolant to the multiple cooling containers at a constant flow rate. In some other examples, the pump may be an active pump, which may be regulated by a controller to vary the flow rate of the coolant supplied to the multiple cooling containers based on temperature of coolant in each of the multiple cooling containers.

(19) Since the electronic devices may not need to be completely submerged in the coolant, a vertically extending rack (e.g., the rack that extends along its height) can be used to stack the electronic devices one on top of other, thereby reducing the floor space required for deploying multiple electronic devices. For example, the rack can utilize the vertical space e.g., up to 50 U or 60 U, hence the floor space utilization is more optimized than the horizontally oriented tank. The ability to use vertically oriented racks to hold the electronic devices may also be facilitated, in part, because in some examples the electronic devices are only partially filled with the coolant. In particular, the usage of cooling containers to define segregated compartments may allow for the overall level of liquid in the coolant pool to be relatively low and thus less coolant is needed, as already described above, and this can greatly reduce the weight of the electronic devices as

compared to an electronic device that is fully filled with coolant. This reduction in weight allows the electronic devices to more easily be disposed in vertically oriented racks, as very heavy electronic devices may be too difficult to install in vertically oriented racks. Further, since the electronic devices are not completely submerged in the coolant and also since the electronic devices are stacked vertically, each electronic device may be partially pulled from the rack, for accessing the electronic components within a corresponding electronic device to perform service and maintenance activities, without shutting down the corresponding electronic device. Further, each electronic device requires less coolant when compared to traditional immersion cooling, thereby reducing the weight of the electronic device and the rack. Additionally, since the existing infrastructure (e.g., the traditional rack and the CDU) can be reused, the cost of deployment of immersion cooling with cooling containers may be significantly reduced when compared to traditional immersion cooling implementations. Accordingly, immersion cooling using cooling containers may allow to significantly improve cooling capabilities of the electronic device, while reducing power utilization of the electronic device.

(20) Referring to the Figures, FIG. 1A depicts a top view of an electronic device **100**. FIG. 1B depicts a side view of the electronic device **100** illustrated while viewing from direction “A”. FIG. 1C depicts another side view of the electronic device **100** illustrated while viewing from the direction “A”. In the description hereinafter, FIGS. 1A-1C are described concurrently for ease of illustration. It should be understood that each of the Figures, FIGS. 1A-1C is not intended to illustrate specific shapes, dimensions, or other structural details accurately or to scale, and that implementations of the electronic device **100** may have different numbers and arrangements of the illustrated components and may also include other parts that are not illustrated.

(21) In some examples, the electronic device **100** may be a computer (e.g., a server, a storage device), a networking device (e.g., a switch), or the like. In the example of FIGS. 1A-1C, the electronic device **100** is a server. The electronic device **100** includes a chassis **102**, a primary circuit board **104**, a plurality of electronic components **106**, a plurality of cooling containers **108**, and a coolant loop **110**.

(22) The chassis **102** includes a pair of lateral sidewalls **114**, a front panel **116**, a rear panel **118**, a base **120**, and a top lid **122**. In such examples, the base **120** is coupled to the pair of lateral sidewalls **114**, the front panel **116**, and the rear panel **118** to define an internal volume **124** of the chassis **102**. Further, the top lid **122** is coupled to the pair of lateral sidewalls **114** and/or to the front panel **116** and the rear panel **118** to define an enclosure. In some examples, the top lid **122** may be configured to be opened or removed from the lateral sidewalls **114** to allow access to the internal volume **124**. For example, the top lid **122** may be pivotably coupled to one of the lateral sidewalls **114** and removably engaged with the lateral sidewalls **114** so that the top lid **122** can be tilted up to open the chassis **102** and provide access to the internal volume **124** of the chassis **102**. As another example, the top lid **122** may be removably attached to the lateral sidewalls by thumbscrews, latches, sliding rails, or any other attachment mechanisms that allow for removal or opening of the top lid **122**. The chassis **102** is partly filled with coolant **125** to form a coolant pool **112** in the internal volume **124** of the chassis **102**. It may be noted that a first lateral sidewall **114A** among the pair of lateral sidewalls **114**, a fifth container **148**, one or more power supply units **136**, a second cooling container **142C**, and one or more memory modules **128** of FIG. 1A are shown to be transparent in FIG. 1B to illustrate other elements of the electronic device **100**. Further, the first lateral sidewall **114A** of FIG. 1A is shown to be transparent in FIG. 1C to illustrate other elements of the electronic device **100**.

(23) In some examples, the base **120**, the pair of lateral sidewalls **114**, the front panel **116**, and the rear panel **118** may be sealed to each other with a liquid-tight seal for preventing leakage of the coolant **125** from the chassis **102**. In other words, in some examples the chassis **102** is a liquid tight enclosure. This may allow the coolant pool **112** to be contained by the chassis **102**. References herein to the lateral sidewalls **114** and/or front/back panels **116**, **118**, and/or chassis **102** being

sealed or liquid tight should be understood as referring to the lateral sidewalls **114** or front/back panels **116**, **118** or chassis being sealed along at least the lower portions thereof (i.e., from the base **120** up to at least a height that exceeds the level of the coolant pool **112**), but such references do not imply that the entire chassis **102** is completely sealed all around unless explicitly indicated otherwise. Thus, for example, a chassis **102** may be referred to as sealed herein even if the chassis **100** includes apertures which are not sealed (e.g., for power and/or communication connectors), so long as those apertures are located higher than the level of the coolant pool **112**. Furthermore, in some examples the top lid **122** is not necessarily sealed to the lateral sidewalls **114** or front/back panels **116**, **118**.

(24) In other examples, an additional structure (not illustrated) having a bottom wall and lateral walls coupled thereto to form a liquid tight cooling container with an open top, is disposed within the chassis **102** on top of the base **120**. This structure may be referred to herein as a tub. The tub may be disposed between the base **120** and some or all of the electronic components of the of the electronic device **100**. In other words, some or all of the electronic components are disposed at least partially in the tub and the tub is disposed in the bottom of the chassis **102**. Moreover, the coolant pool **112** may be disposed in the tub. The tub may serve as a way to hold the coolant **125** within the electronic device **100** without necessarily requiring that the chassis **102** be liquid tight. Or the chassis **102** may be liquid tight and yet still include the tub as a form of redundant protection against leaks. In some examples that include such a tub, the tub might not extend fully across the entirety of the base **102**, and thus the coolant pool **112** might not fully cover the entirety of the base **102**.

(25) In one or more examples, the coolant **125** may be a thermally conductive dielectric coolant. Examples of such coolants include mineral oil, synthetic oil, hydrocarbon fluid, or the like.

(26) The primary circuit board **104** is disposed within and coupled to the base **120** of the chassis **102** (directly, or via an intermediate structure such as the tub described above). In one or more examples, at least part of the primary circuit board **104** is disposed in the coolant pool **112** and immersed in the coolant **125**. In particular, a level of the coolant **125** in the coolant pool **112** may be high enough to cover at least part of the primary circuit board **104** and electronic elements (e.g., resistors, capacitors, or the like) disposed on and coupled to a top side of the primary circuit board **104**. For example, in some implementations the entirety of the primary circuit board **104** is disposed in the coolant pool **112** excluding any portions that may be contained within cooling containers **108** (described below). In some examples, the primary circuit board **104** may be a motherboard of the electronic device **100**.

(27) The electronic components **106** are disposed within the chassis **102**. In some examples, the electronic components **106** may occupy some portions of the internal volume **124** of the chassis **102**, when disposed within the chassis **102**. In certain examples, the electronic components **106** includes components of the electronic device **100** such as one or more central processing units (CPUs **126**), one or more memory modules **128** (e.g., DIMMS), one or more storage devices **130** (e.g., solid state drives), one or more expansion cards **132** (e.g., PCIe cards), one or more network cards **134** (OCP cards), one or more power supply units **136**, and the like. In some examples, some of the electronic components **106** are coupled to the primary circuitry board **104** while other electronic components **106** are not. For example, the storage devices **130** or other electronic components may be coupled to another circuit board **131** (often called a backplane, midplane, or mezzanine). In some examples, the electronic components **106** may be grouped into a first subset **106-1** of the plurality of electronic components **106** and a second subset **106-2** of the plurality of electronic components **106**. In some examples, the first subset **106-1** includes the electronic components **106** that are mounted on the primary circuit board **104** while the second subset **106-2** includes the electronic components **106** not mounted on the primary circuit board **104**. In certain examples, the first subset **106-1** includes the one or more CPUs **126**, the one or more memory modules **128**, the one or more expansion cards **132**, and the one or more network cards **134**. In

certain examples, the second subset **106-2** includes the one or more storage devices **130** and the one or more power supply units **136**. For example, the one or more storage devices **130** and the one or more power supply units **136** are mounted on and coupled to the base **120** (directly, or via an intermediate structure, such as circuit board **131**). Even though in the example of FIGS. **1A-1C**, the one or more storage devices **130** and the one or more power supply units **136** are shown to be disposed on the base **120** of the chassis **100**, in some other examples, the one or more storage devices **130** and the one or more power supply units **136** may be mounted on the primary circuit board **104**. In some examples, the electronic device **100** further includes one or more external communication ports **138** (e.g., USB ports, network ports, or the like), which are mounted to one of the panels, e.g., the rear panel **118** of the chassis **102**. In particular, the one or more external communication ports **138** may be coupled to yet another circuit board **139**, which may be coupled to the printed circuit board **104** via cables (not shown), for example. In certain examples, each of the one or more external communication ports **138** is disposed at a height that is greater than a level of the coolant **125** in the coolant pool **112** to prevent leakage of the coolant **125** through the one or more external communication ports **138**.

(28) The plurality of cooling containers **108** is disposed within the chassis **102** such that each cooling container among the plurality of cooling containers **108** contains at least one electronic component among the plurality of electronic components **106**. In some examples, the cooling containers **108** may also occupy some portions of the internal volume **124** of the chassis **102**, when disposed within the chassis **102**.

(29) In certain examples, the cooling containers **108** includes multiple different types of cooling containers **108** that have different shapes and/or sizes and that may enclose different components **106**. Some of the cooling containers **108** may include lateral walls coupled to the primary circuit board **104**, with a portion of the primary circuit board **104** forming the bottom bound of the cooling container. Other cooling containers **108** may include lateral walls coupled to the bottom **120** of the chassis **102** (or to an intermediate structure, such as a tub), with the bottom **120** (or intermediate structure) forming the bottom bound of the cooling container. Each cooling container **108** may include at least one inlet and at least one outlet, wherein coolant flows into the cooling container **108** via the inlet and out of the cooling container via the outlet. In some examples, the outlet of one cooling container **108** forms the inlet of another cooling container **108**. In some cases, the inlet/outlet may include an aperture or guide feature, such as a port, a valve, a chute, a pipe, etc., while in other cases the outlet may be a top of a wall of the cooling container **108** (e.g., the coolant leaves the cooling container **108** by overflowing the wall) and the inlet of the cooling container **108** may be a top opening of the cooling container **108**. At least one of the cooling containers **108** may be configured to receive the coolant from a coolant loop, described in greater detail below. In some examples, one or more of the cooling containers **108** are configured to receive coolant from the outlet of another one of the cooling container **108**.

(30) For example, in the implementation illustrated in FIGS. **1A-1C**, the cooling containers **107** include a plurality of first cooling containers **140**, a plurality of second cooling containers **142**, a third cooling container **144**, a fourth cooling container **146**, and a fifth cooling container **148**. In some examples, the plurality of cooling containers **108** may be grouped into a first set of cooling containers **108-1** and a second set of cooling containers **108-2**. In the example of FIGS. **1A-1C**, the first set of cooling containers **108-1** includes the plurality of first cooling containers **140**, the plurality of second cooling containers **142**, and the third cooling container **144**. Similarly, the second set of cooling containers **108-2** includes the fourth cooling container **146** and the fifth cooling container **148**. In some examples, each of the first set of cooling containers **108-1** is disposed on the primary circuit board **104** and contains at least one electronic component of the first subset **106-1** of the plurality of electronic components **106**. For example, the plurality of first cooling containers **140** contains the one or more CPUs **126**, the plurality of second cooling containers **142** contains the one or more memory modules **128**, and the third cooling container **144**

contains the one or more expansion cards **132** and the one or more network cards **134**. Further, each cooling container among the second set of cooling containers **108-2** is not disposed on the primary circuit board **104** and contains at least one electronic component of the second subset **106-2** of the plurality of electronic components **106**. In particular, each cooling container among the second set of cooling containers **108-2** is disposed on the base **120** and contains the at least one electronic component of the second subset **106-2**. For example, the fourth cooling container **146** is disposed on and coupled to the base **120** such that the fourth cooling container **146** contains the one or more storage devices **130**. Similarly, the fifth cooling container **148** is disposed on and coupled to the base **120** such that the fifth cooling container **148** contains the one or more power supply units **136**.

(31) In certain examples, the plurality of first cooling containers **140** includes a first cooling container **140A** and another first cooling container **140B**. In such examples, the first cooling container **140A** includes an inlet e.g., a first inlet **150A**, and one or more outlets e.g., a first outlet **152A**, and a second outlet **152B**. Similarly, the other first cooling container **140B** includes another inlet e.g., a second inlet **150B**, and one or more outlets e.g., a third outlet **152C** and fourth outlet **152D**. In certain examples, the plurality of second cooling containers **142** includes a second cooling container **142A**, another second cooling container **142B**, and yet another second cooling container **142C**. In such examples, the second cooling container **142A** includes a fifth outlet **154A**. However, the first outlet **152A** of the first cooling container **140A** may function as an inlet of the second cooling container **142A**. Similarly, the other second cooling container **142B** includes a sixth outlet **154B** and a seventh outlet **154C**. However, the second outlet **152B** of the first cooling container **140A** and the third outlet **152C** of the other first cooling container **140B** function as inlets of the second cooling container **142B**. Further, the other second cooling container **142C** includes an eighth outlet **154D**. However, the fourth outlet **152D** of the other first cooling container **140B** functions as an inlet of the second cooling container **142C**. In certain examples, the third cooling container **144** includes a third inlet **156** and one or more outlets e.g., a ninth outlet **158A**, a tenth outlet **158B**, and an eleventh outlet **158C**. In certain examples, the fourth cooling container **146** includes a fourth inlet **160** and one or more outlets e.g., a twelfth outlet **162A**, a thirteenth outlet **162B**, and a fourteenth outlet **162C**. In certain examples, the fifth cooling container **148** includes a fifth inlet **164** and one or more outlets e.g., a fifteenth outlet **166A** and a sixteenth outlet **166B**. In one or more examples, each of the first outlet **152A**, the second outlet **152B**, the third outlet **152C**, and the fourth outlet **152D** is configured to overflow the coolant **125** into a corresponding second cooling container **142A**, **142B**, and **142C**. However, each of the fifth outlet **154A**, the sixth outlet **154B**, the seventh outlet **154C**, the eighth outlet **154D**, the ninth outlet **158A**, the tenth outlet **158B**, the eleventh outlet **158C**, the twelfth outlet **162A**, the thirteenth outlet **162B**, the fourteenth outlet **162C**, the fifteenth outlet **166A**, and the sixteenth outlet **166B** is configured to overflow the coolant **125** into the coolant pool **112** of the chassis **102**. Even though not illustrated, in some other examples, each of the plurality of second cooling containers **142** may be separately coupled to the manifold **178** via one or more pipes, instead of fluidically coupled to one of the first containers **140** to receive the coolant **125** overflowing from the corresponding first container **140**. In such examples, each of the first cooling containers **140** may also overflow the coolant **125** directly to the coolant pool **112** instead of overflowing the coolant **125** to the second container **142**.

(32) The heights of each cooling container **108** may be configured based on the dimension so the components **106** enclosed thereby so that the coolant level in the cooling container is sufficiently height to cover at least the portions of the component **106** that need cooling (which may include the entirety of the component **106** in some cases, or may include only a portion of the component **106** in other cases). Because the various components may have different heights, the cooling containers **108** too may have different heights and different coolant levels therein. In one or more examples, each of the plurality of cooling containers **108** has a height greater than a height of a corresponding electronic component **106**, so as to allow each cooling container **108** to immerse the corresponding electronic component **106** in the coolant **125** filled within the corresponding cooling container **108**.

In other examples, portions of some of the components **106** may protrude above the level of the coolant **125**, such as handles or other portions that do not need direct cooling. In some examples, at least the electrically active portions of the electronic component **106** may be covered by the coolant **125**. Further, at least one cooling container among the plurality of cooling containers **108** has a height that is different than at least one other cooling container among the plurality of cooling containers **108**. In other words, the height of each cooling container **108** is defined to be greater than that of the height of the corresponding electronic component **106**. In particular, referring to FIG. **1B**, the third cooling container **144** has the height which is greater than the height of each cooling container of the one or more first cooling containers **140** and the height of the fourth cooling container **146**. Further, the third cooling container **144** has a height which is greater than the height of the one or more expansion cards **132** and the one or more network cards **134**. Similarly, referring to FIG. **1C**, each cooling container of the one or more second cooling containers **142** has the height which is smaller than the height of a corresponding cooling container of the one or more first cooling containers **140**. Further, the second cooling container **142** has a height which is greater than the height of the one or more memory modules **128**. Similarly, the fifth cooling container **148** has a height which is greater than the height of a corresponding cooling container of the one or more second cooling containers **142**. Further, the fifth cooling container **148** has a height which is greater than the height of the one or more power supply units **136**.

(33) The one or more cooling containers **108** among the first set of cooling containers **108-2** are disposed on the primary circuit board **104**. In some examples, each cooling container **108** among the first set of cooling containers **108-1** has vertical wall portions **168** defining lateral bounds of the cooling container **108**. In such examples, the vertical wall portions **168** are sealed liquid tight to the primary circuit board **104** such that the primary circuit board **104** defines a bottom bound of the cooling container **108**. The first cooling container **140** has first vertical wall portions **168A**, which are sealed liquid tight to the primary circuit board **104**. In such examples, the first vertical wall portions **168A** define the lateral bounds and a portion of the primary circuit board **104** that is configured to mount the one or more CPUs **126** to the primary circuit board **104** defines the bottom bound of the first cooling container **140**. Accordingly, the first vertical wall portions **168A** and the portion of the primary circuit board **104** are sealed liquid tight to hold the coolant **125** filled within the first cooling container **140** and prevent leakage of the coolant **125** from the first cooling container **140** to the coolant pool **112** via the primary circuit board **104**. The second cooling container **142** has second vertical wall portions **168B**, which are sealed liquid tight to the primary circuit board **104**. In such examples, the second vertical wall portions **168B** define the lateral bounds and a portion of the primary circuit board **104** that is configured to mount the one or more memory modules **128** to the primary circuit board **104** define the bottom bound of the second cooling container **142**. Accordingly, the first vertical wall portions **168B** and the portion of the primary circuit board **104** are sealed liquid tight to hold the coolant **125** filled within the second cooling container **142** and prevent leakage of the coolant **125** from the second cooling container **142** to the coolant pool **112** via the primary circuit board **104**. The third cooling container **144** has third vertical wall portions **168C**, which are sealed liquid tight to the primary circuit board **104** (directly through a portion of the primary circuit board **104** and through a horizontal wall portions **168D**). In such examples, the third vertical wall portions **168C** define the lateral bounds and the horizontal wall portion **168D** and the portion of the primary circuit board **104** define the bottom bounds of the third cooling container **144**. Accordingly, the third vertical wall portions **168C**, the portion of the primary circuit board **104**, and the horizontal wall portion **168D** are sealed liquid tight to hold the coolant **125** filled within the third cooling container **144** and prevent leakage of the coolant **125** from the third cooling container **144** to the coolant pool **112** via the primary circuit board **104**. Even though each of the first cooling container **140**, the second cooling container **142**, and the third cooling container **144** is shown to have lower bounds sealed liquid tight to the primary circuit board **104** to hold the coolant **125** filled within the corresponding container, in

some other examples, each of the first cooling container **140**, the second cooling container **142**, and the third cooling container **144** may have the lower bounds not sealed liquid tight to the primary circuit board **104**, thereby allowing a portion of the coolant **125** to leak therethrough from the corresponding container to the coolant pool **112**. In such examples, the supply of the coolant **125** to the corresponding container may be maintained at a higher rate than the leakage rate of the coolant **125** to maintain sufficient amount of coolant **125** in the corresponding container or prevent draining out of the corresponding container. In some other examples, the lower bounds of each of each of the first cooling container **140**, the second cooling container **142**, and the third cooling container **144** may be coupled to the base **120** of the chassis **102**.

(34) The one or more cooling containers **108** among the second set of cooling containers **108-2** are disposed on the base **120** of the chassis **102**. In particular, the fourth cooling container **146** is disposed on the base **120** such that the fourth cooling container **146** contains the one or more storage devices **130**. Further, the fourth cooling container **146** may be coupled to the base **120**. Similarly, the fifth cooling container **148** is disposed on the base **120** such that the fifth cooling container **148** contains the one or more power supply units **136**. Further, the fifth cooling container **148** may be coupled to the base **120**.

(35) In one or more examples, the coolant loop **110** is configured to circulate the coolant **125** from the coolant pool **112** to the plurality of cooling containers **108**. In some examples, the coolant loop **110** includes a coolant collector **170**, one or more circulation pipes **172**, a heat exchanger **174**, a pump **176**, a manifold **178**, and one or more distribution pipes e.g., pipes **180**.

(36) The coolant collector **170** (as shown in FIG. 1A) is disposed within the coolant pool **112**. The coolant collector **170** may be disposed such that an opening therein through which coolant **125** enters the coolant collector **170** is at a lower height than the coolant level (i.e., the top surface of the coolant) in the coolant pool **112** to allow the coolant collector **170** to collect the coolant **125** from the coolant pool **112** and supply the collected coolant **125** to the heat exchanger **174**. The heat exchanger **174** is fluidically connected to the coolant collector **170** and the pump **176** via the one or more circulation pipes **172**. The heat exchanger **174** is further fluidically connected to a facility water (not shown) via an inlet tube **184** and an outlet tube **186**. In some examples, the heat exchanger **174** is an indirect heat exchanger, which may receive the relatively warm coolant **125** from the coolant collector **170** via one of the one or more circulation pipes **172** and relatively cool facility liquid **182** from the inlet tube **184**. Further, the heat exchanger **174** may transfer heat from the coolant **125** into the facility liquid **182**, thus cooling the coolant **125** and warming the facility liquid **182**. Later, the heat exchanger **174** may output the now-cooled coolant **125** via another circulation pipe **172** of the one or more circulation pipes **172** and output the now-warmed facility liquid **182** via the outlet tube **186**. The cooled coolant **125** is then supplied to the cooling containers **108** via the remainder of the coolant loop **110**. The pump **176** generates the pressure which causes the coolant **125** to flow in this manner.

(37) In the illustrated example, the pump **176** is located downstream of the heat exchanger **174** and upstream of the manifold **178**. In particular, in the illustrated example, the pump **176** is fluidically connected to the heat exchanger **174** via the circulation pipe **172** to which the heat exchanger **174** outputs the cooled coolant **125**. The pump **176** is further fluidically connected to the manifold **178** via yet another circulation pipe **172** of the one or more circulation pipes **172**. The pump **176** drives the coolant **125** to flow from the heat exchanger **174** to the manifold **178**. In other examples the pump **176** is located in other positions in the loop **110**. For example, the pump **176** may be upstream of the heat exchanger **174** (e.g., between the heat exchanger **174** and the coolant collector **170**).

(38) The manifold **178** is fluidically connected to at least one of the cooling containers **108** via at least one pipe **180** to distribute the coolant **125** to the plurality of cooling containers **108**. In some examples, the manifold **178** is fluidically connected to multiple cooling containers **108** via multiple pipes **180**. In some examples, one or more cooling containers **108** are not directly coupled to the

manifold **178** via pipes **180**, and instead such cooling containers **108** may receive an inflow of coolant **125** from another one of the cooling container **108** (i.e., an outflow of coolant **125** from one cooling container **108** becomes the inflow of coolant **125** to another cooling container **108**).

(39) In some examples, the plurality of pipes **180** includes a first pipe **180A**, a second pipe **180B**, a third pipe **180C**, a fourth pipe **180D**, and a fifth pipe **180E**. In one or more examples, the first pipe **180A** extends from the manifold **178** and connects to the first inlet **150A** of the first cooling container **140A**. Similarly, the second pipe **180B** extends from the manifold **178** and connects to the second inlet **150B** of the other first cooling container **140B**. The third pipe **180C** extends from the manifold **178** and connects to the third inlet **156** of the third cooling container **144**. The fourth pipe **180D** extends from the manifold **178** and connects to the fourth inlet **160** of the fourth cooling container **146**. Further, the fifth pipe **180E** extends from the manifold **178** and connects to the fifth inlet **164** of the fifth cooling container **148**. In some examples, respective diameters of the plurality of pipes **180** vary depending on the respective cooling container size and levels of the coolant **125** in corresponding cooling containers **108** to which the plurality of pipes **180** are connected, respectively, among the one or more cooling containers **108**. In particular, each of the first pipe **180A** and the second pipe **180B** has diameter which may be greater than the diameter of the third pipe **180C**, the fourth pipe **180D**, and the fifth pipe **180E** to allow the coolant **125** to fill in the corresponding cooling container **108**, when the pump **176** is operating at a constant flow rate.

(40) In one or more examples, the first cooling container **140A** includes the first inlet **150A** fluidically coupled to the coolant loop **110** to receive an inflow of a first portion **125A** of the coolant **125** from the coolant loop **110**. Similarly, the other first cooling container **140B** includes the second inlet **150B** fluidically coupled to the coolant loop **110** to receive the inflow of a second portion **125B** of the coolant **125** from the coolant loop **110**. The third cooling container **144** includes the third inlet **156** fluidically coupled to the coolant loop **110** to receive the inflow of a third portion **125C** of the coolant **125** from the coolant loop **110**. The fourth cooling container **146** includes the fourth inlet **160** fluidically coupled to the coolant loop **110** to receive the inflow of a fourth portion **125D** of the coolant **125** from the coolant loop **110**. Finally, fifth cooling container **148** includes the fifth inlet **164** fluidically coupled to the coolant loop **110** to receive the inflow of a fifth portion **125E** of the coolant **125** from the coolant loop **110**. In such examples, the second cooling container **142A** is configured to receive a part of the outflow of the coolant **125** overflowing from the first cooling container **140A** as the inflow of the coolant **125** of the second cooling container **142A**. In such examples, the other second cooling container **142B** is configured to receive another part of the outflow of the coolant **125** overflowing from the first cooling container **140A** as the inflow of the coolant **125** of the other second cooling container **142B**. Similarly, the other second cooling container **142C** is configured to receive a part of the outflow of the coolant **125** overflowing from the other first cooling container **140B** as the inflow of the coolant **125** of the other second cooling container **142C**. In such examples, the other second cooling container **142B** is configured to receive yet another part of the outflow of the coolant **125** overflowing from the other first cooling container **140B** as the inflow of the coolant **125** of the other second cooling container **142B**. Further, the outflow of the coolant from each of the second cooling containers **142A**, **142B**, **142C** overflows into the coolant pool **112** in the chassis **102**. Similarly, the outflow of the coolant from each of the third cooling container **144**, the fourth cooling container **146**, and the fifth cooling container **148** overflows into the coolant pool **112** in the chassis **102**.

(41) During operation of the electronic device **100**, in certain examples, the pump **176** may operate at a constant flow rate. In such examples, the pump **176** may draw the coolant **125** from the coolant collector **170** into the manifold **178** via the heat exchanger **174**. In particular, when the pump **176** drives the coolant **125** from the coolant collector **170**, the coolant **125** may first flow through the heat exchanger **174**, where the heat exchanger **174** may use the facility liquid **182** to indirectly remove heat from the coolant **125**. Further, the coolant **125** may flow from the heat exchanger **174**

to the manifold **178**.

(42) In some examples, the manifold **178** may split the coolant **125** into multiple portions. In such examples, the first cooling container **140A** may receive the inflow of the first portion **125A** of the coolant **125** from the manifold **178**. The first portion **125A** of the coolant **125** may raise up to the height of the first outlet **152A** of the first cooling container **140A** and remove heat from the one or more CPUs **126**. The second cooling container **142A** may receive a part **125A.sub.1** of the outflow of the coolant **125** overflowing from the first cooling container **140A** as the inflow of the coolant **125** of the second cooling container **142A**. The part **125A.sub.1** of the coolant **125** may raise up to the height of the fifth outlet **154A** of the second cooling container **142A** and remove heat from the one or more memory modules **128**. Further, the outflow of the part **125A.sub.1** of the coolant **125** of the second cooling container **142A** may overflow through the fifth outlet **154A** into the coolant pool **112** in the chassis **102**.

(43) Similarly, the other first cooling container **140B** may receive the inflow of the second portion **125B** of the coolant **125** from the manifold **178**. The second portion **125B** of the coolant **125** may raise up to the height of the fourth outlet **152D** of the second cooling container **140B** and remove heat from the one or more CPUs **126**. The second cooling container **142C** may receive a part **125B.sub.1** of the outflow of the coolant **125** overflowing from the first cooling container **140B** as the inflow of the coolant **125** of the second cooling container **142C**. The part **125B.sub.1** of the coolant **125** may raise up to the height of the eighth outlet **154D** of the second cooling container **142C** and remove heat from the one or more memory modules **128**. Further, the outflow of the part **125B.sub.1** of the coolant **125** of the second cooling container **142C** may overflow through the eighth outlet **154D** into the coolant pool **112** in the chassis **102**.

(44) In one or more examples, the second cooling container **142B** may receive other parts **125A.sub.2**, **125B.sub.2** of the coolant **125** from the first cooling container **140A** and the other first cooling container **140B**. The other parts **125A.sub.2**, **125B.sub.2** of the coolant **125** may raise up to the height of the sixth outlet **154B** and the seventh outlet **154C** of the second cooling container **142B** and remove heat from the one or more memory modules **128**. Further, the outflow of the other parts **125A.sub.2**, **B.sub.2** of the coolant **125** of the second cooling container **142B** may overflow through the sixth outlet **154B** and the seventh outlet **154C** into the coolant pool **112** in the chassis **102**. In other words, the outflow of the coolant **125** of the second cooling container overflows into the coolant pool **112** in the chassis **102**.

(45) The third cooling container **144** may receive the third portion **125C** of the coolant **125** from the manifold **178**. The third portion **125C** of the coolant **125** may raise up to the height of the ninth outlet **158A**, the tenth outlet **158B** and the eleventh outlet **158C** of the third cooling container **144** and remove heat from the one or more expansion cards **132** and the one or more network cards **134**. Further, the outflow of the third portion **125C** of the coolant **125** of the third cooling container **144** may overflow through the ninth outlet **158A**, the tenth outlet **158B** and the eleventh outlet **158C** into the coolant pool **112** in the chassis **102**. The fourth cooling container **146** may receive the fourth portion **125D** of the coolant **125** from the manifold **178**. The fourth portion **125D** of the coolant **125** may raise up to the height of the twelfth outlet **162A**, the thirteenth outlet **162B**, and the fourteenth outlet **162C** of the fourth cooling container **146** and remove heat from the one or more storage devices **130**. Further, the outflow of the fourth portion **125D** of the coolant **125** of the fourth cooling container **146** may overflow through the twelfth outlet **162A**, the thirteenth outlet **162B**, and the fourteenth outlet **162C** of into the coolant pool **112** in the chassis **102**. The fifth cooling container **148** may receive the fifth portion **125E** of the coolant **125** from the manifold **178**. The fifth portion **125E** of the coolant **125** may raise to the height of the fifteenth outlet **166A** and the sixteenth outlet **166B** of the fifth cooling container **148** and remove heat from the one or more power supply units **136**. Further, the outflow of the fifth portion **125E** of the coolant **125** of the fifth cooling container **148** may overflow through the fifteenth outlet **166A** and the sixteenth outlet **166B** into the coolant pool **112** in the chassis **102**. In one or more examples, the coolant **125** (e.g.,

125A.sub.1, 125B.sub.1, 125A.sub.2B.sub.2, 125C, 125D, 125E) in the coolant pool 112 may remove heat from the primary circuit board 104 and flow into the coolant collector 170. The pump may drive the coolant 125 in the coolant collector 170 to the manifold 178 via the heat exchanger 174 and the process of recirculating the coolant 125 may continue as discussed herein.

(46) FIG. 2 depicts a top view of one implementation example of the electronic device 100 of FIGS. 1A-1C having a controller 202 and a plurality of sensors 204. In some examples, each of the plurality of sensors 204 is a temperature sensor disposed within a respective cooling container of a plurality of cooling containers 108. In particular, each of the plurality of sensors 204 is disposed such that it contacts the coolant 125 filled in the respective cooling container and detects the temperature of the coolant 125 filled in that cooling container 108. The plurality of sensors 204 includes a plurality of first sensors 204A, a plurality of second sensors 204B, a third sensor 204C, a fourth sensor 204D, and a fifth sensor 204E. In such examples, one sensor among the plurality of first sensors 204A is disposed in the first cooling container 140A and another sensor among the plurality of first sensors 204A is disposed in the other first cooling container 140B. Similarly, one sensor among the plurality of second sensors 204B is disposed in the second cooling container 142A, another sensor among the plurality of second sensors 204B is disposed in the other second cooling container 142B, and yet another sensor among the plurality of second sensors 204B is disposed in the other second cooling container 142C. The third sensor 204C is disposed in the third cooling container 144, the fourth sensor 204D is disposed in the fourth cooling container 146, and the fifth sensor 204E is disposed in the fifth cooling container 148. Further, each of the plurality of sensors 206 is communicatively coupled to the controller 202. For example, each of the plurality of first sensors 204A is communicatively coupled to the controller 202 via a respective first input signal 206A. Similarly, each of the plurality of second sensors 204B is communicatively coupled to the controller 202 via a respective second input signal 206B. The third sensor 204C is communicatively coupled to the controller 202 via a third input signal 206C. The fourth sensor 204D is communicatively coupled to the controller 202 via a fourth input signal 206D. The fifth sensor 204E is communicatively coupled to the controller 202 via a fifth input signal 206E.

(47) In some examples, the controller 202 may be hosted in the electronic device 100. In some other examples, the controller 202 is deployed in outside of the electronic device 100. The controller 202 may have memory to store a threshold temperature data corresponding to each of the plurality of electronic components 106. For example, the threshold temperature data may include a first threshold temperature data corresponding to the one or more CPUs 126, a second threshold temperature data corresponding to the one or more memory modules 128, a third threshold temperature data corresponding to the one or more expansion cards 132 and the one or more network cards 134, a fourth threshold temperature data corresponding to the one or more storage devices 130, and a fifth threshold temperature data corresponding to the one or more power supply units 136. In one or more examples, the threshold temperature data may be a temperature data indicative of a maximum temperature up to which a corresponding electronic component 106 may operate without deteriorating a performance of the corresponding electronic component 106. The controller 202 may be communicatively coupled to the pump 176 via a control signal 208. In such examples, the pump 176 is an active pump, which may be regulated to vary a flow rate of the coolant 125 based on the control signal 208 received from the controller 202.

(48) During operation, each of the plurality of sensors 204 may periodically (e.g., at a time interval of about 30 seconds, for example) detect the temperature of the corresponding electronic component 106 by sensing the temperature of the coolant 125 filled in the corresponding cooling container 108. In other words, each sensor detects temperature of the coolant 125 in the corresponding cooling container 108. Further, each of the plurality of sensors 204 may send a detected temperature data of each of the plurality of electronic components 106 via the corresponding input signal (206A, 206B, 206C, 206D, 206E) to the controller 202. In such examples, the controller 202 may receive the detected temperature data of each of the plurality of

electronic components **106** and compare the detected temperature data with a corresponding threshold temperature data of each of the plurality of electronic components **106**. In some examples, when the controller **202** determines that the detected temperature data of one of the plurality of electronic components **106** is greater than the corresponding threshold temperature data, the controller **202** sends the control signal **208** to the pump **176** to vary (e.g., increase) the flow rate of the coolant **125** to the plurality of cooling container **108**. In other words, the controller **202** generates the control signal **208** to the pump **176** based on a comparison of the detected temperature data with the threshold temperature data. In such examples, the pump **176** drives the coolant **125** to the plurality of cooling containers **108** at a varied flow rate based on the control signal **208** received from the controller **202**.

(49) FIG. 3 depicts a block diagram of a rack **300** having an electronic device **100** of FIGS. 1A-1C and a facility liquid supply unit **302**. It should be understood that each of the FIG. 3 is not intended to illustrate specific shapes, dimensions, or other structural details accurately or to scale, and that implementations of the rack **300** may have different numbers and arrangements of the illustrated components and may also include other parts that are not illustrated. The rack **300** includes a facility liquid supply unit **302**, a plurality of enclosures **304**, and one or more electronic devices **100**. In some examples, the rack **300** may be a vertically extending cabinet having at least a 42 Unit (U) height, 22 U height, 36 U height, or the like.

(50) The plurality of enclosures **304** includes a first enclosure **304A**, a second enclosure **304B**, a third enclosure **304C**, a fourth enclosure **304D**, and a fifth enclosure **304E** located vertically spaced apart from each other. In some examples, the facility liquid supply unit **302** is a coolant distribution unit. In some examples, the facility liquid supply unit **302** is disposed in the first enclosure **304A** among the plurality of enclosures **304**. The facility liquid supply unit **302** includes an inlet tube **184** and an outlet tube **186**. The inlet tube **184** may be configured to connect to an inlet side of a heat exchanger **174** of each of the one or more electronic devices **100**. Similarly, the outlet tube **186** may be configured to connect to an outlet side of the heat exchanger **174** of each of the one or more electronic devices **100**. The inlet tube **184** may supply the facility liquid (e.g., cool facility liquid) to the inlet side of the heat exchanger **174** and the outlet tube **186** may receive the facility liquid **182** (e.g., a hot facility liquid) from the outlet side of the heat exchanger **174**. In such examples, the facility liquid supply unit **302** may further include a facility pump **306** and a heat dissipater **308** coupled to the inlet tube **184** and the outlet tube **186**. The heat dissipater **308** may receive the facility liquid **182** from the outlet tube, remove heat from the facility liquid **182**, and supply the facility liquid **182** to the facility pump **306**. In such examples, the facility pump **306** may drive the facility liquid **182** to the heat exchanger **174** of each of the one or more electronic device **100**.

(51) The one or more electronic devices **100** includes a first electronic device **100A** and a second electronic device **100B**. Each of the one or more electronic devices **100** e.g., the first electronic device **100A** may have a removable top lid **122** disposed on and coupled to the chassis **102**. The first electronic device **100A** is disposed in the second enclosure **304B** and the second electronic device **100B** is disposed in the third enclosure **304C**. The fourth and fifth enclosures **304D**, **304E** are shown empty in the example of FIG. 3. Each of the first and second electronic devices **100A**, **100B** is similar to the electronic device **100** discussed in the example of FIGS. 1A-1C, therefore, each of the first and second electronic devices **100A**, **100B** is not discussed herein for brevity of the disclosure. In one or more examples, each of the one or more electronic devices **100** may be slidably inserted into the corresponding enclosure **304** of the rack **300**. In such examples, in a state of the electronic device, e.g., the first electronic device **100A** being partially pulled out from the second enclosure **304B**, the removable top lid **122** may provide access to the plurality of electronic components **106** (as shown in FIGS. 1A-1C) disposed within the chassis **102** for performing a service or maintenance event.

(52) In some examples, the heat exchanger **174** of the first electronic device **100A** and the second electronic device (not shown) is disposed in a thermal contact with the inlet tube **184** and the outlet

tube **186** of the facility liquid supply unit **302**. The heat exchanger **174** is configured to remove heat from the coolant **125** by indirectly exchanging heat between the facility liquid **182** and the coolant **125**. In such examples, the pump (not shown) of the one or more electronic devices **100** drives the coolant **125** to the plurality of cooling containers **108** (as shown and discussed in the example of FIGS. **1A-1C** and **3**). Further, the facility liquid **182** (e.g., hot facility liquid) is driven to the outlet tube **186** of the facility liquid supply unit **302**.

(53) FIG. **4** depicts a flowchart depicting a method **400** of immersion cooling an electronic device. It may be noted herein that the method **400** is described in conjunction with FIGS. **1A-1C**, and **3** for example. The method **400** starts at block **402** and continues to block **404**.

(54) At block **404**, the method **400** includes circulating a coolant from a coolant pool to a plurality of cooling containers via a coolant loop. In some examples, the coolant pool is formed from the coolant that is partly filled in an internal volume of a chassis of an electronic device. The coolant loop includes a pump, which is configured to drive the coolant from a coolant collector to the plurality of cooling containers via a heat exchanger, a manifold, and a corresponding pipe connected to the manifold. In some examples, the heat exchanger removes heat from the coolant and the pump outputs the coolant to the manifold. The method **400** continues to block **406**.

(55) At block **406**, the method **400** includes receiving an inflow of the coolant by each cooling container among the plurality of cooling containers from the coolant loop or from another one of the plurality of cooling containers, and immerse at least one electronic component among a plurality of electronic components of the electronic device in the coolant to remove heat from the at least one electronic component, wherein the plurality of cooling containers is disposed within the chassis such that each cooling container contains the at least one electronic component. In some examples, the manifold bifurcates the coolant received from the pump into multiple portions. In such examples, some of the plurality of cooling containers may directly receive a portion of coolant among the multiple portions of coolant from the manifold. However, some adjacent cooling containers may receive the coolant from some of the plurality of cooling containers. In particular, the coolant from some of the plurality of cooling containers may overflow into some adjacent cooling containers. The method **400** continues to block **408**.

(56) At block **408**, the method **400** includes outputting an outflow of the coolant from each cooling container that overflows into the coolant pool or into at least one adjacent cooling container of the plurality of cooling containers, wherein a level of the coolant in the coolant pool in the chassis is lower than respective levels of the coolant in the plurality of cooling containers. In some examples, some of the plurality of cooling containers may overflow the coolant into the coolant pool.

However, some other cooling containers of the plurality of cooling containers may overflow the coolant into the adjacent cooling containers. The coolant in such adjacent cooling containers may remove the heat from the at least one electronic component and may overflow into the coolant pool. The coolant in the coolant pool may immerse a primary circuit board to remove heat from the primary circuit board. The coolant may later flow into the coolant collector and the circulation of the coolant may happen again to remove the heat from the plurality of electronic components of the electronic device, as discussed herein. The method **400** ends at block **410**.

(57) Since the electronic devices may not need to be completely submerged in the coolant, a vertically extending rack (e.g., the rack that extends along its height) can be used to stack the electronic devices one on top of other, thereby reducing the floor space required for deploying multiple electronic devices. For example, the rack can utilize the vertical space up to 50 U or 60 U, hence the floor space utilization is more optimized than the horizontally oriented rack. Further, since the electronic devices are not completely immersed in the coolant and also since the electronic devices are stacked vertically, each electronic device may be partially pulled from the rack, for accessing the electronic components within a corresponding electronic device and perform service and maintenance activity, without shutting down the corresponding electronic device. Also, each electronic device may require less coolant when compared to traditional immersion cooling,

thereby reducing the weight of the electronic device and the rack. Further, since the existing infrastructure (e.g., the rack and the CDU) can be reused, the cost of deployment of the immersion cooling may be significantly reduced. Additionally, since the first cooling container that contains high-powered electronic components receives the coolant directly from the coolant loop and the second cooling container that contains low-powered electronic components receives the coolant from the first cooling container, the cooling and power usage effectiveness of each electronic device can be significantly improved.

(58) In the foregoing description, numerous details are set forth to provide an understanding of the subject matter disclosed herein. However, implementation may be practiced without some or all of these details. Other implementations may include modifications, combinations, and variations from the details discussed above. It is intended that the following claims cover such modifications and variations.

Claims

1. An electronic device comprising: a chassis having an internal volume configured to be partly filled with a coolant to form a coolant pool in the internal volume; a plurality of electronic components disposed within the chassis; a plurality of cooling containers disposed within the chassis such that each cooling container among the plurality of cooling containers contains at least one electronic component among the plurality of electronic components; and a coolant loop configured to circulate the coolant from the coolant pool to the plurality of cooling containers, wherein each cooling container among the plurality of cooling containers is configured to: receive an inflow of the coolant from the coolant loop or from another one of the plurality of cooling containers and immerse the at least one electronic component in the coolant to remove heat from the at least one electronic component, and output an outflow of the coolant that overflows into the coolant pool in the chassis or into at least one adjacent cooling container of the plurality of cooling containers, wherein a level of the coolant in the coolant pool in the chassis is lower than respective levels of the coolant in the plurality of cooling containers.
2. The electronic device of claim 1, wherein the coolant loop comprises a pump, a heat exchanger, a manifold disposed within the chassis, and a coolant collector disposed within the coolant pool, wherein the heat exchanger is configured to receive the coolant from the coolant pool via the coolant collector, remove heat from the coolant, and output the coolant to the manifold, and wherein the pump is configured to drive the coolant to flow from the coolant collector to the plurality of cooling containers via the heat exchanger and the manifold.
3. The electronic device of claim 2, wherein the pump drives the coolant to the plurality of cooling containers at a constant flow rate.
4. The electronic device of claim 2, further comprising a controller and a plurality of temperature sensors, where each sensor is disposed in a corresponding cooling container among the plurality of cooling containers, wherein the controller is communicatively coupled to the plurality of temperature sensors and the pump, wherein each sensor detects temperature of the coolant in the corresponding cooling container and communicates detected temperature data to the controller, wherein the controller generates a control signal to the pump based on a comparison of the detected temperature data with a threshold temperature data, and wherein the pump drives the coolant to the plurality of cooling containers at a varied flow rate based on the control signal received from the controller.
5. The electronic device of claim 1, wherein the chassis is a liquid tight enclosure.
6. The electronic device of claim 1, wherein at least one cooling container among the plurality of cooling containers has a height that is different than at least one other cooling container among the plurality of cooling containers.
7. The electronic device of claim 1, further comprising: a primary circuit board, wherein a first

subset of the plurality of electronic components are mounted on the primary circuit board and a second subset of the plurality of electronic components are not mounted on the primary circuit board; wherein the plurality of cooling containers comprises a first set of cooling containers and a second set of cooling containers, wherein each cooling container among the first set of cooling containers is disposed on the primary circuit board and contains at least one electronic component of the first subset, and wherein each cooling container among the second set of cooling containers is not disposed on the primary circuit board and contains at least one electronic component of the second subset.

8. The electronic device of claim 7, wherein each cooling container among the first set of cooling containers comprises vertical wall portions defining a lateral bounds of the cooling container, wherein the vertical wall portions are sealed liquid tight to the primary circuit board such that the primary circuit board defines a bottom bound of the cooling container.

9. The electronic device of claim 7, wherein one or more cooling containers among the second set of cooling containers are disposed on a base of the chassis.

10. The electronic device of claim 7, wherein at least part of the primary circuit board is disposed in the coolant pool in the internal volume of the chassis and immersed in the coolant to remove heat from the primary circuit board.

11. The electronic device of claim 7, wherein the first set of cooling containers comprises a first cooling container and a second cooling container, wherein the first cooling container comprises an inlet fluidically coupled to the coolant loop to receive the inflow of the coolant from the coolant loop, wherein the second cooling container is configured to receive a part of the outflow of the coolant overflowing from the first cooling container as the inflow of the coolant of the second cooling container, and wherein the outflow of the coolant of the second cooling container overflows into the coolant pool in the chassis.

12. The electronic device of claim 11, wherein the first cooling container comprises one or more outlets configured to guide the outflow of the coolant to the second cooling container.

13. The electronic device of claim 11, wherein the first set of cooling containers further comprises another second cooling container, wherein the other second cooling container is configured to receive another part of the outflow of the coolant overflowing from the first cooling container as the inflow of the coolant of the other second cooling container.

14. The electronic device of claim 13, wherein the first cooling container comprises a first outlet configured to guide the part of the outflow of the coolant to the second cooling container and a second outlet configured to guide the other part of the outflow of the coolant to the other second cooling container.

15. The electronic device of claim 1, further comprising one or more external communication ports disposed on a panel of the chassis, wherein the external communication port is disposed at a height greater than the level of the coolant in the coolant pool in the chassis.

16. The electronic device of claim 1, wherein one or more cooling containers among the plurality of cooling containers are fluidically connected to a pump via a manifold and a corresponding pipe among a plurality of pipes connected to the manifold, wherein respective diameters of the plurality of pipes vary depending on the respective levels of the coolant in corresponding cooling containers to which the plurality of pipes are connected, respectively, among the one or more cooling containers.

17. A rack comprising: a plurality of enclosures; a facility liquid supply unit disposed along a first enclosure among the plurality of enclosures, wherein the facility liquid supply unit comprises an inlet tube and an outlet tube; and the electronic device of claim 1, disposed in a second enclosure among the plurality of enclosures, where a heat exchanger of the electronic device is disposed in a thermal contact with the inlet tube and the outlet tube of the facility liquid supply unit to remove heat from the coolant.

18. The rack of claim 17, wherein the electronic device of claim 1, further comprising a removable

top lid disposed on and coupled to the chassis, wherein, in a state of the electronic device being partially pulled out from the second enclosure, the removable top lid provides access to the plurality of electronic components disposed within the chassis.

19. A method comprising: circulating a coolant from a coolant pool to a plurality of cooling containers via a coolant loop, wherein the coolant pool is formed from the coolant that is partly filled in an internal volume of a chassis of an electronic device; receiving an inflow of the coolant by each cooling container among the plurality of cooling containers from the coolant loop or from another one of the plurality of cooling containers, and immerse at least one electronic component among a plurality of electronic components of the electronic device in the coolant to remove heat from the at least one electronic component, wherein the plurality of cooling containers is disposed within the chassis such that each cooling container contains the at least one electronic component; and outputting an outflow of the coolant from each cooling container that overflows into the coolant pool or into at least one adjacent cooling container of the plurality of cooling containers, wherein a level of the coolant in the coolant pool in the chassis is lower than respective levels of the coolant in the plurality of cooling containers.

20. The method of claim 19, wherein circulating the coolant via the coolant loop comprises: receiving the coolant from a coolant collector disposed within the coolant pool by a heat exchanger; removing heat from the coolant, and outputting the coolant to a manifold; and driving the coolant to flow from the coolant collector to the plurality of cooling containers by a pump via the heat exchanger and the manifold.
